

# Communication Tone, Pull Request Outcomes, and Contributor Return in Open Source

## Part 3: Final Results

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# The Question

## Core question (observational)

Do communication signals in pull request discussion associate with whether a PR gets merged and whether the author comes back?

RQ1 Which signals associate with **PR merge outcome**?

RQ2 Which signals associate with **author return** at 30 / 90 / 180 days?

RQ3 Are **newcomers** affected differently from experienced contributors?

RQ4 Does **politeness** buffer the effect of isolated harsh language?

## Signals measured per PR

Toxicity (max), sentiment, anger — transformer models | Politeness — lexical heuristic

# The Dataset

## Scale

- **12,098** focal PRs (Jan 2024 – Mar 2025)
- **3** repositories: Kubernetes, Next.js, scikit-learn
- **2,210** PR authors; **100k+** scored comments

## Key rates (mean $\pm$ SD)

- Merge: **75.7%**    Return 30d: **76.8%**
- Return 90d: **81.3%**    180d: **83.5%**
- Prior PR count: mean 65.7, SD 96.1

## Key descriptive fact

Only **1.0%** of PRs have any comment with toxicity  $\geq 0.05$ . These are well-moderated projects.

## Data leakage prevention

A **6-month lookback** (Jul–Dec 2023) precedes the focal window. This ensures contributor history is available before any focal PR is observed — preventing leakage into newcomer classification.

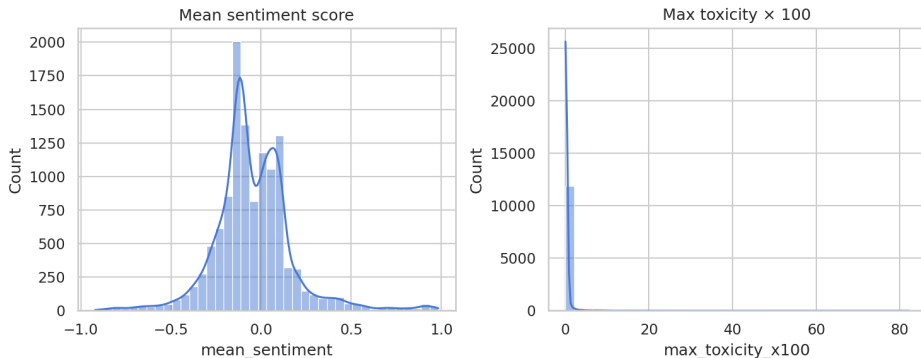
## Variable types and model choice

- Outcomes (*merged*, *returned*) are **binary** → logistic regression
- NLP signals are **continuous**  $\in [0, 1]$  or  $[-1, 1]$
- Controls: prior PR count, comment count, participants, response delay, repo fixed effects
- HC3 robust standard errors; VIF 1.01–1.25 (no multicollinearity)

## Key cleaning decisions

- Bots filtered by login pattern
- Comments < minimum length discarded
- Code blocks and URLs stripped before NLP scoring
- Only **received** comments scored (excludes PR author's own text)
- Duplicate comment texts deduplicated before model inference

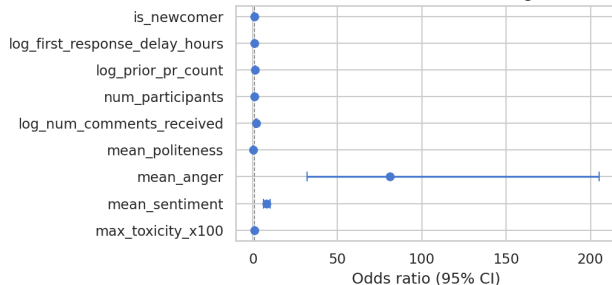
# What the Signals Look Like



- **Sentiment** is well-distributed across PRs — a usable continuous signal
- **Max toxicity** is extremely right-skewed: almost all PRs sit near zero
- This is why averaging toxicity fails and why max toxicity and binary flags are needed

# RQ1: What Predicts Merge?

RQ5: Odds ratios for PR merge



$N = 12,098$ , McFadden pseudo- $R^2 = 0.118$

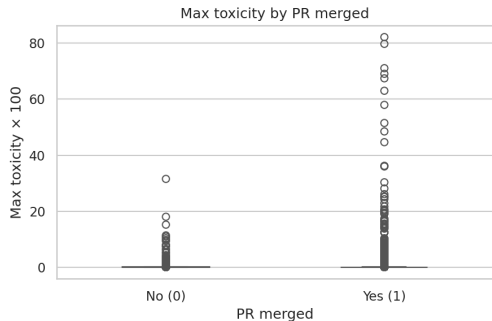
## Significant signals

- **Sentiment** strongly positive  
coef = +2.10,  $p < 10^{-60}$
- **Anger** positive  
coef = +4.40,  $p < 10^{-20}$
- **Politeness** negative  
coef = -1.42,  $p < 10^{-12}$

## Note

Anger and politeness direction explained on next slide.

# Why the Anger and Politeness Findings Make Sense



Merged PRs carry the high-toxicity outliers — raw data confirms the direction.

## Anger → more merging

“This is completely broken” scores as *anger* in a social-media model. In code review it signals **engaged review** pushing toward a decision.

## Politeness → fewer merges

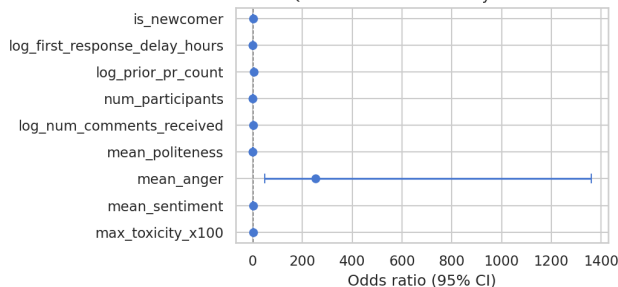
“Maybe consider...”, “could you perhaps...” marks **soft rejection**, not approval.

## Key lesson

NLP signals carry real information, but not what they were designed to capture. Domain context is essential.

## RQ2: What Predicts Return?

RQ1: Odds ratios for 90-day retention



$N = 12,098$ , McFadden pseudo- $R^2 = 0.498$

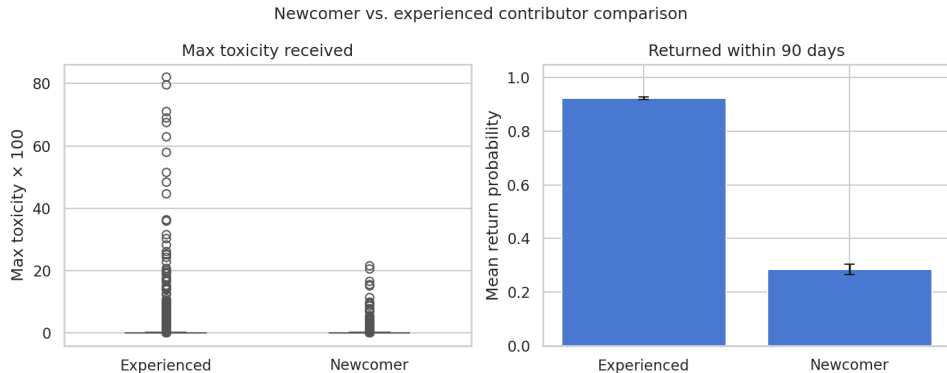
### Significant signals

- **Prior PR count dominates**  
coef = +1.51,  $p < 10^{-179}$
- **Anger still positive**  
coef = +5.54,  $p < 10^{-10}$
- Toxicity, sentiment, politeness:  
**not significant**

### Main finding

Return is about **who you already are**,  
not the tone of one PR.

## RQ3: Newcomers vs. Experienced Contributors



- **Left:** toxicity exposure is similar for both groups — newcomers are not treated more harshly
- **Right:** experienced contributors return at  $\sim 92\%$ ; newcomers at  $\sim 28\%$
- Newcomer  $\times$  toxicity interaction: **not significant** ( $p = 0.57$ ) — the gap is about

## RQ4: Politeness as a Buffer

Does a polite discussion soften isolated harsh comments?

Interaction term:  $\text{max\_toxicity} \times \text{mean\_politeness}$  in the 90-day return model.

- Interaction is **significant**:  $p = 0.003$ ,  $\text{coef} = +0.81$
- When overall discussion is polite, isolated harsh comments associate *less* negatively with return
- The overall culture of the discussion absorbs individual sharp moments

### Interpretation

Context matters: one sharp comment in an otherwise civil discussion lands differently than the same comment in a hostile thread. Community tone is a cushion.

# What We Learned

## From the results

- Communication tone matters more for **merge** than for **return**
- Return is dominated by **contributor experience** — one bad discussion does not override years of history
- Overt toxicity is too rare in large moderated repos to drive retention outcomes on its own
- The politeness buffer suggests that **overall community culture** absorbs individual harsh moments

## From the method

NLP signals trained on social media do not map cleanly to code review. Technical assertiveness scores as anger; tentative rejection scores as politeness. Domain context is essential for interpretation.

## Broader takeaway

The biggest risk to losing contributors is not any single PR discussion — it is failing to onboard them deeply enough to begin with.

## Design limitations

- Observational: associations only, not causes
- Return = opening another PR, not all activity
- Only 3 large English-language repositories
- NLP models trained on social media, not code review
- Author fixed effects failed (singular matrix)

## Robustness checks

- Binary toxicity  $\geq 0.05$ : retention OR significant ( $p < 0.001$ )
- Binary toxicity  $\geq 0.10$ : not significant ( $p = 0.15$ )
- Results change with threshold — toxicity is sparse and operationalization-sensitive

### On toxicity

The data do **not** support a simple “toxic comments drive people away” story.

# Experience & Reflections

## What I learned

- Running a full empirical pipeline end-to-end: API collection, NLP scoring, regression analysis
- Why data leakage matters — the lookback window was a non-obvious design decision that changed the results
- Off-the-shelf NLP models carry domain assumptions that can **invert** your expected findings

## What was hardest

- Toxicity is so rare (1% of PRs) that standard analysis almost cannot find it
- Operationalization choices — mean vs. max, which threshold — changed significance
- Deciding what counts as a *result* vs. an artifact of the model

## If I did it again

I would validate the NLP signals against domain-specific ground truth (e.g., manual annotation of code review tone) before trusting their direction

| Question                   | Finding                            | Strength    |
|----------------------------|------------------------------------|-------------|
| RQ1: signals and merge     | sentiment +, anger +, politeness – | strong      |
| RQ2: signals and return    | prior experience dominates         | very strong |
| RQ3: newcomer sensitivity  | no differential toxicity effect    | null result |
| RQ4: politeness moderation | buffers isolated harsh comments    | moderate    |

## One-sentence conclusion

Communication tone shapes **whether your PR gets accepted**, but **who you already are** determines whether you come back.

Questions?



*Repository*